

(KTXFI2EBNF) Physics II. Exam 1 Solutions
December 16, 2025

I. Moving Reference Frames. Special Theory of Relativity

1. Write down the Lorentz transformation equations used in the special theory of relativity. Specify the exact physical meaning of the symbols used in the formulas, including the reference frame in which they are defined. (5 points)

Lecture on October 13, 2025, Slides 18–20.

2. Write down Einstein's relation expressing mass–energy equivalence. What is its physical meaning? (5 points)

Lecture on October 13, 2025, Slides 23–24.

II. Limits of the Classical Conceptual Framework

3. What is the photoelectric effect? Explain the mechanism of the photoelectric effect. (5 points)

Lecture on October 6, 2025, Slides 23–27.

4. Describe the Compton experiment and provide a qualitative but precise explanation of the phenomenon of Compton scattering. (5 points)

Lecture on October 6, 2025, Slides 29–31.

5. Formulate the de Broglie principle in your own words. (5 points)

Lecture on October 20, 2025, Slide 23.

III. Elements of Quantum Mechanics. Physics of Condensed Matter. Theory of Electrical Conduction

6. What does the Heisenberg uncertainty principle express? To which types of physical quantity pairs does it apply? (5 points)

Lecture on October 13, 2025, Slides 31–35.

7. What is the Peltier effect and what is its explanation? Draw a diagram. Interpret the effect and provide the physical explanation of the phenomenon. (5 points)

Lecture on November 10, 2025, Slides 19–21.

IV. Development of Atomic Theory

8. State the Pauli exclusion principle. (5 points)

Lecture on October 20, 2025, Slide 18.

9. Give the physical interpretation of the principal quantum number. What is the principal quantum number? What does it represent? What is its physical meaning? What values can it take? (5 points)

Lecture on October 20, 2025, Slide 14.

10. Given the element with atomic number 10 in the periodic table, write down its electronic configuration using the standard atomic physics notation. (5 points)

Lecture on October 20, 2025, Slides 19–22.

Grades: 0–25 points: Fail (1), 26–30 points: Pass (2), 31–37 points: Satisfactory (3), 38–45 points: Good (4), 46–50 points: Excellent (5).

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